

Irene Liu

liuirene256@gmail.com | <https://liuirene256.github.io/> | [linkedin.com/in/liuirene256](https://www.linkedin.com/in/liuirene256)

EDUCATION

Carnegie Mellon University | Pittsburgh, PA

May 2026

Bachelor of Science in Computer Science | **Minor in Computational Finance**

GPA: 3.95/4.0 (Dean's List High Honors, all semesters)

Relevant Courses: Algorithm Design & Analysis, Systems Design, Parallel Algorithms & Data Structures, Functional Programming, Theoretical Computer Science, Machine Learning, Continuous Math Finance, Discrete Math Finance, Probability for Computing, Math Foundations for CS, Linear Algebra (Honors)

WORK EXPERIENCE

Sentry | San Francisco, CA

Sept 2025 – Dec 2025

Software Developer Intern

- Foundations Team (Incoming Fall 2025)

Five Rings | New York, NY

June 2025 – Aug 2025

Software Developer Intern

- Engineered a **momentum trading bot** in C++ for the proprietary trading simulation, enabling strategy validation
- Ensured reliability by writing comprehensive **unit tests** for order placement/cancellation, and message processing
- Safeguarded the core trading infrastructure by **refactoring the middleware dispatcher** against adversarial messages

Magnetar Capital | Evanston, IL

June 2024 – July 2024

Software Engineering Intern

- Developed program to dynamically profile 700+ directories and notify stakeholders of missing market files
- Achieved infrastructure stability by enhancing in-house **Kubernetes deployment** with monitoring interface
- Reduced back-office overtime toil by **streamlining alert communications**, leading to an 80% decrease in latency

Panduit Corporation | Tinley Park, IL

May 2024 – June 2024

Firmware Engineering Intern

- Enhanced network transmission between SNMP agents and data center infrastructure for 600+ database objects
- Pioneered feature to accommodate multiple administrative users, **increasing revenue by 15%**
- Decreased runtime by 95% through **caching** of data and **parallelization** of database retrievals or modifications

Carnegie Mellon University | Pittsburgh, PA

Jan 2024 – June 2025

Teaching Assistant

- Teaching Assistant for 15-151: Mathematical Foundation in Computer Science (aka Concepts)
- Teaching Assistant for 21-241: Linear Algebra
- Teaching Assistant for 21-266: Vector Calculus for Computer Scientists

PROJECTS

ML Nasal Disease Diagnostic Software · Python

AI – 2023

- Increases diagnosis accessibility by leveraging an SVM and **signal processing algorithms** to classify speech
- Achieves a 12% increase in accuracy and 99% decrease in cost compared to PROM-based clinician diagnosis

SKILLS

Proficient Languages: Python, Java, C/C++, C#, SML, HTML/CSS, JavaScript, jQuery, SQL, MATLAB

Tools: Git, Vim, Docker, Kubernetes, Linux, AWS, CMake, GDB, LaTeX, MongoDB, TensorFlow, Figma, MS Office

Other: Machine Learning, Data Analysis, Databases, Unit Testing, Test Automation, Networking Protocols, Mathematics

AWARDS

AIME Qualifier, Hedging Contest t10, best MongoDB Hack, ACSL 1st Place State, National Merit Scholar, National Debater